

A noise-scaling window for concentration of a shared flow

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Abstract

Uniform microscopic activity need not give every flow channel equal disturbance: channels coupled to more interactions can collect more fluctuations. We connect this mechanism to a conditional prediction for concentrated sharing of a fixed total flow. For reflected driftless dynamics with flow weight g^m and variance rate proportional to $g^q / \sum_j g_j^m$, we derive the exact joint stationary law. The population fraction above the pooled half-flow threshold vanishes as both state range and population grow precisely when $1 \leq q \leq m + 1$, for exact powers. At fixed population N , it instead approaches $1/(2N)$ inside that window. Independent interaction contributions proportional to actual flow recover $q = m$; existing deterministic lattice examples demonstrate seeding and flow-dependent variation, without establishing the complete diffusion limit or both boundaries. The window concerns scaling across channel states, not overall noise amplitude. We state finite-range physical tests, including an amplitude-rescaling null. The power-law moment threshold is established mathematically; the contribution is its exact conserved-flow formulation and testable physical interpretation. Physical applicability remains untested.

1 The physical prediction

Can the same microscopic activity, acting throughout a substrate, sort a fixed flow into a small fraction of large-share channels? A route coupled to more microscopic interactions can collect more perturbations under the same local rule. This paper asks when that state-dependent disturbance permits extreme concentration. It addresses sharing among available routes; it does not derive the formation of those routes.

The proposed prediction is conditional: physical systems with the flow and fluctuation laws specified below should follow the finite-range concentration curve, with asymptotic separation only in the scaling window $1 \leq q \leq m + 1$. This is more specific than the statement that disturbance helps channels grow. Outside the window, a vanishing population fraction does not carry half the pooled flow as the accessible range grows. Unequal sharing can nevertheless remain.

The distinction between scaling and amount is essential. After accounting for the shared normalization, doubling channel state multiplies flow weight by 2^m and variance rate by 2^q . The exponent q concerns variance, not the size of a typical kick, which scales with its square root. Increasing activity equally at every interaction multiplies all variance rates by a constant. In this model that speeds up evolution without changing the stationary distribution. It cannot by itself traverse the window. The physical question is how coupling to microscopic activity and its correlations make disturbance increase across channel states.

Noise-dependent residence times and stationary occupation have a long history [1, 2]. Power-weighting a power-law distribution gives a standard moment threshold; Newman discusses both this condition and the fraction of a weighted total held by a population tail [3]. These ingredients are not claimed as new. Shared-flow normalization also changes the dynamical clock. Making it explicit gives the exact joint law, finite-population correction and observable needed to assess this physical mechanism.

2 A model with conservation imposed at every instant

Let $N \geq 1$ channels have dimensionless states $g_i \in [1, R]$, where $R > 1$. The model specifies possible flow routes and their response law; it does not prescribe which routes become dominant. The deterministic example in Sec. 3 tests differentiation from identical coarse-route states. Spatial formation and connectivity of routes in a material are abstracted out: no spatial energy-gradient field or changing network topology is modeled here. For $m > 0$, define the total weight S and the actual flow through channel i by

$$S(\mathbf{g}) = \sum_{j=1}^N g_j^m, \quad J_i(\mathbf{g}) = I \frac{g_i^m}{S(\mathbf{g})}. \quad (1)$$

Thus $\sum_i J_i = I$ for every configuration, not only in expectation. We choose the interior dynamics

$$dg_i = \sqrt{2D_i(\mathbf{g})} dW_i, \quad D_i(\mathbf{g}) = \frac{N g_i^q}{S(\mathbf{g})}, \quad (2)$$

with independent Wiener processes, normal reflection at both endpoints, and zero interior drift. Boundary local-time terms are implicit. A positive common prefactor has been absorbed into the time unit. The lower cutoff sets the state unit; all powers refer to this specified coordinate.

The normalization in Eq. (2) is a constitutive assumption. On the line $q = m$, $D_i = (N/I)J_i$, so noise variance rate is proportional to actual flow. Away from that line the model keeps the common factor $1/S$. The apparently similar choice $D_i \propto J_i^{q/m}$ instead contains $S^{-q/m}$ and defines different dynamics. Conservation of flow alone does not select either rule.

The Itô convention would follow from an appropriate small-kick limit of conditionally unbiased additive events in g , evaluated at the pre-event state. Such a limit also needs control of temporal correlations and higher moments. Unresolved deterministic dynamics do not automatically satisfy these conditions. Under a nonlinear change $y = h(g)$, Itô's formula adds the drift $D(g)h''(g)$ even when the drift in g is zero. The zero-drift assumption must therefore be justified in the measured coordinate, rather than restored after arbitrary relabelling.

3 From uniform microscopic activity to differential disturbance

Let each microscopic contribution be conditionally unbiased, with the same variance $v dt$ in the specified state coordinate. For n_i independent contributions, variance addition gives $2D_i = n_i v$. More generally,

$$\text{Var}\left(\sum_{a=1}^{n_i} \xi_a\right) = \sum_a \text{Var}(\xi_a) + 2 \sum_{a < b} \text{Cov}(\xi_a, \xi_b). \quad (3)$$

Thus uniform local activity can produce unequal coarse disturbance through interaction number and correlations. Here n_i is an effective count of coupled contributions, not a surface area or a spatial dimension. The lattice can index unresolved degrees of freedom rather than physical positions. A spatial contact footprint is one realization; the event construction below instead uses interaction rates. Statistical uniformity means equal microscopic rules and statistics, not identical states or synchronized trajectories.

If the coupling obeys $n_i = MJ_i/I = Mg_i^m/S$, independence gives $D_i = (Mv/2)g_i^m/S$. A common choice of time unit recovers Eq. (2) with $q = m$. Here differential noise follows from the interaction rule rather than being specified separately for each channel. This line lies in the concentration window for $m \geq 1$; it lies below it for $0 < m < 1$. More interactions alone are therefore not a universal sufficient condition.

One explicit small-kick construction gives the same connection. At total rate $2N/h^2$, an identical interaction event chooses route i with probability J_i/I and adds $+h$ or $-h$ with equal

probability; attempted outward moves are rejected. Each allowed signed jump then has rate $N g_i^m / (S h^2)$, the lattice approximation to the $q = m$ diffusion. Event placement implements the flow-dependent coupling. That coupling is a constitutive assumption; the variance scaling then follows without specifying channel geometry. This identifies the target coarse generator; its event statistics still need to be recovered from the deterministic substrate.

The repository's deterministic footprint example instead sums fluctuations from chaotic patches with fixed coupling per patch and footprint proportional to flow share. It gives a variance-versus-mean-share slope of 1.01, consistent with independent variance addition. A separate deterministic example starts forty coarse routes identically: three fine-state seeds yield final top-three shares .615, .463 and .529, against .075 initially and in frozen controls. No random numbers are drawn during evolution. The latter example prescribes a share-dependent amplitude, so it is a seeding demonstration rather than an independent derivation of contact growth. A periodic control also concentrates; chaos is not shown necessary.

These examples establish parts of the mechanism, not a complete match to the bounded diffusion. Their regressions use route-wise time averages; their boundaries and temporal correlations differ. Deterministic diffusion limits have established precedents [5, 6], but require conditions not verified here. Perfectly coherent contributions would give variance proportional to n_i^2 ; with $n_i \propto J_i$, that also changes the shared denominator to S^2 . It cannot be substituted into this theorem by changing q alone. A microscopic construction crossing both boundaries under the same normalization remains open.

4 Exact stationary population and flow distributions

On $[1, R]$, write

$$Z_R = \int_1^R g^{-q} dg, \quad p_0(g) = \frac{g^{-q}}{Z_R}, \quad \mu_R = \int_1^R g^m p_0(g) dg, \quad f_R(g) = \frac{g^m p_0(g)}{\mu_R}. \quad (4)$$

All quantities are finite at finite R , for every real q .

Proposition 1. *The unique stationary density of Eq. (2) on the reflecting box is*

$$\pi_{N,R}(\mathbf{g}) = \frac{S(\mathbf{g})}{N \mu_R} \prod_{j=1}^N p_0(g_j). \quad (5)$$

Its tagged-channel marginal and its pooled flow density are, respectively,

$$p_{N,R}(g) = \left(1 - \frac{1}{N}\right) p_0(g) + \frac{1}{N} f_R(g), \quad (6)$$

$$\mathbb{E}_\pi \left[\sum_{i=1}^N \frac{J_i}{I} \delta(g - g_i) \right] = f_R(g). \quad (7)$$

Proof. Integration of $S \prod_j p_0(g_j)$ gives $N \mu_R$, establishing normalization. For each coordinate,

$$D_i \pi_{N,R} = \frac{1}{Z_R \mu_R} \prod_{j \neq i} p_0(g_j)$$

is independent of g_i . Every probability current $-\partial_i(D_i \pi_{N,R})$ therefore vanishes, including at the boundaries. The diffusion is uniformly nondegenerate on the connected compact box, giving uniqueness. Expanding the sum S and integrating out the other coordinates gives Eq. (6). Multiplication by $J_i/I = g_i^m/S$ cancels S in the joint law; integration and summation give Eq. (7). $\square$

The calculation also has a simple clock interpretation. If $L_0 = \sum_i g_i^q \partial_i^2$ is the generator of independent reflected channels, the coupled generator is $L = (N/S)L_0$. Configurations with larger S are traversed more slowly relative to that independent process; their stationary weight acquires the factor S . This explains both the solvability and the dependence of the result on the chosen normalization.

An exact independent sampling construction is useful: choose one channel uniformly, draw its state from f_R , and draw the other $N - 1$ states independently from p_0 . Averaging over the chosen channel reproduces Eq. (5). That designation is resampled for each configuration; it does not identify a permanent winner.

The correction matters even in simple cases. For $N = 1$ and $q = m$, the channel carries all the flow and has constant diffusivity. Its stationary density is uniform, not g^{-m} . At fixed R , the tagged marginal approaches p_0 as $N \rightarrow \infty$. The independent-channel calculation is recovered in that limit.

5 The concentration observable and its window

Let a_R be the state threshold above which half of the pooled flow lies:

$$\int_{a_R}^R f_R(g) dg = \frac{1}{2}. \quad (8)$$

Define $P_0(R) = \int_{a_R}^R p_0(g) dg$ and $P_N(R) = \int_{a_R}^R p_{N,R}(g) dg$. Equation (6) gives

$$\boxed{P_N(R) = \left(1 - \frac{1}{N}\right) P_0(R) + \frac{1}{2N}.} \quad (9)$$

This counts a population fraction above one threshold fixed across stationary observations. It is a pooled observable, with each observation weighted equally in physical time.

Theorem 1. *For Eqs. (1)–(2) with exact powers and $m > 0$, $P_N(R) \rightarrow 0$ as both $R \rightarrow \infty$ and $N \rightarrow \infty$ if and only if*

$$\boxed{1 \leq q \leq m + 1.} \quad (10)$$

At fixed N , the limit inside this window is $1/(2N)$.

Proof. Put $A = m - q + 1$ and $B = 1 - q$. Direct integration gives

$$a_R = \begin{cases} [(R^A + 1)/2]^{1/A}, & A \neq 0, \\ \sqrt{R}, & A = 0, \end{cases} \quad P_0(R) = \begin{cases} (R^B - a_R^B)/(R^B - 1), & B \neq 0, \\ \log(R/a_R)/\log R, & B = 0. \end{cases} \quad (11)$$

For $q < 1$, $a_R/R \rightarrow 2^{-1/A}$ and $P_0 \rightarrow 1 - 2^{-B/A} > 0$. At $q = 1$, $P_0 \sim \log 2/(m \log R) \rightarrow 0$. For $1 < q < m + 1$, a_R grows proportionally to R , while the normalized population tail above it vanishes. At $q = m + 1$, $P_0 = 1/(R^{m/2} + 1) \rightarrow 0$. For $q > m + 1$, $a_R \rightarrow 2^{1/(q-m-1)}$, so $P_0 \rightarrow 2^{-(q-1)/(q-m-1)} > 0$. Equation (9) supplies the finite- N result and the joint limit. $\square$

Either iterated limit gives this classification, as does any sequence with both R and N tending to infinity. Stationarity is taken at each bounded pair (N, R) first. No unbounded diffusion or uniform equilibration time is asserted. The same classification holds for any fixed pooled flow fraction strictly between zero and one, with the corresponding fraction divided by N replacing $1/(2N)$.

The integrals behind the two edges are $\int g^{-q} dg$ and $\int g^{m-q} dg$. At the lower edge the population normalization is logarithmic; at the upper edge the flow normalization is logarithmic. The upper-edge population fraction itself decays as a power of R . Figure 1 displays the finite-range expressions.

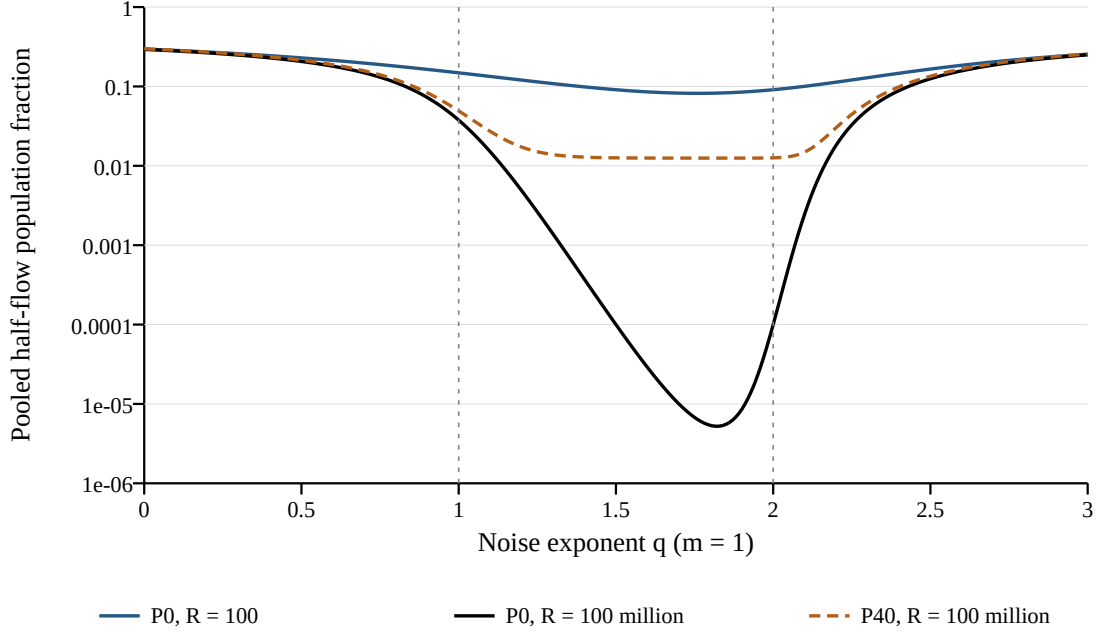


Figure 1: Exact stationary predictions, not empirical measurements or large-range dynamic simulations. The asymptotic window is between the dotted lines. Finite range smooths both boundaries; finite population supplies a floor. P_0 is the large-population limit at each fixed R . The curve at $R = 10^8$ is not the infinite-range limit.

5.1 Pooled concentration is not a snapshot winner count

In a single configuration, rank shares in descending order and count the smallest number of whole channels whose shares sum to at least one half. Call that count divided by N H_N . It always satisfies $H_N \geq 1/N$. In contrast, P_N can approach $1/(2N)$ because its fixed threshold need not carry half the flow in every snapshot. These observables must not be substituted for one another. The numerical checks measure $\mathbb{E}H_N$ separately against the exact joint sampler. Neither statistic measures how long a particular channel remains dominant. Stationary concentration alone establishes neither persistent structure nor Darwinian selection.

For example, at $m = 1$, $q = 2$, $R = 10^8$, $P_0 \simeq 0.000100$, whereas $P_{40} \simeq 0.012598$. This is a substantial relative correction to the tail fraction, even though $1/N$ is small.

5.2 Limiting slopes do not determine the endpoints

The inclusive endpoints require exact powers or additional tail assumptions. On $g \geq e$, consider the unnormalized flow weight $w(g) = g$ and noise factor $d(g) = g^2(\log g)^2$. Their limiting log-log slopes are $m = 1$ and $q = 2$, but $\int_e^\infty w(g)/d(g)dg$ converges. The upper-edge separation fails. This example also fits the coupled construction with $S = \sum_i w(g_i)$ and $D_i = Nd(g_i)/S$: replace p_0 by the normalized $1/d$ and f_R by the normalized w/d in the proof. Logarithmic corrections do not change the strict interior power inequalities, but can decide the boundary cases. An estimated slope near an edge is consequently insufficient to classify a measured system.

6 What would make the result physically useful?

The upper bound excludes a proposed mechanism when the measured disturbance scaling is too steep relative to the flow weight, provided the rest of the model applies. It also distinguishes

state-dependent disturbance from monotonic reinforcement intuition. At $m = 1$, the large-range, large-population half-flow fraction vanishes for $q = 2$, but approaches $1/8$ for $q = 2.5$. Above the boundary it can still be very small, especially close to $q = 2$. The mathematical transition is the loss of a vanishing fraction, not an abrupt return to equal sharing in a finite apparatus.

The basic inequality is a standard power-law moment condition [3]. Its value here is the connection to actual conserved shares and the observable correction in Eq. (9). These make a conditional prediction that can be checked. They do not establish a new general law of nature, and no priority claim for the elementary threshold is made.

A fair physical test needs simultaneous channel states and individual flows, with a fixed total and a stable channel population. It must estimate both conditional drift and variance rate in the chosen coordinate, check cross-channel correlations, and allow enough relaxation before comparing stationary statistics. For Eq. (2), conditioning on the configuration or accounting for S is necessary: a regression on g_i alone need not recover q . Drift, measurement noise, jumps, changing populations and limited state range can each prevent the test from addressing this model. Power-law fits themselves also need checks against alternatives [4].

The discriminating test has two parts. First, vary a common activity prefactor at fixed interaction law: after adequate relaxation, the stationary curve should remain unchanged. Second, vary the state dependence of interaction-driven disturbance, while verifying the same shared normalization, and compare the finite-range curve across the upper edge. Fit the dynamical parameters without fitting the concentration outcome; predict that outcome on separate observations. Change R and, where feasible, N to distinguish a finite-range effect from the asymptotic classification. No finite apparatus should exhibit an exact singular boundary.

A failed model-assumption check means that this experiment does not test this particular prediction. Conversely, disagreement with the finite-range curve after independent checks of those assumptions would count against the proposed model. No physical dataset is used as validation in this paper.

7 Numerical checks of the stationary result

The repository implements a continuous-time nearest-neighbour chain on a uniform state lattice with spacing h . Each allowed jump of channel i by $+h$ or $-h$ has rate

$$\lambda_i(\mathbf{g}) = \frac{Ng_i^q}{S(\mathbf{g})h^2}. \quad (12)$$

Outward boundary jumps are absent. Exponential holding times and rate-weighted event selection simulate this finite chain without a time-step approximation. The lattice invariant law is exactly Eq. (5) with sums replacing integrals. Grid refinement remains necessary for a continuous-state comparison. Observations are taken at fixed physical times, because sampling only jump events would bias residence-time statistics.

Eight automated tests include detailed balance on every edge of an enumerated three-channel, three-state system; independent marginal calculations; the single-channel case; the finite-population quantile identity; sampling checks; continuum integral comparisons; grid refinement; and relaxation from both boundaries in a simple control. The retained ensemble runs use six exponent pairs: $(1, 0.5)$, $(1, 1)$, $(1, 1.5)$, $(1, 2)$, $(1, 2.5)$ and $(2, 4)$. For each pair they use lower-boundary, upper-boundary, log-uniform and exact stationary initial conditions, with three seeds and 128 independent ensembles per seed. Physical observation times are 0, 1, 4, 16.

All retained final errors are below the declared tolerance of 0.05 (Table 1). This is an acceptance criterion, not a confidence interval. Separate exact-law calculations check grid refinement and evaluate large state ranges; they are not dynamical runs at those ranges. The

N	Grid intervals	Occupancy CDF	Flow CDF	Mean snapshot fraction
8	4	0.017014	0.021730	0.011423
8	8	0.020299	0.024502	0.005979
40	8	0.009496	0.008539	0.001585

Table 1: Largest final absolute errors across the six exponent pairs and four initial conditions for each retained run, all on $[1, 3]$. CDF errors compare with the exact lattice law. The snapshot reference uses 100,000 independent joint-law samples per pair. Seeds are pooled at the final observation time.

checks support the implementation on the tested bounded grids. They do not establish large-range equilibration times or empirical applicability.

The legacy independent-walker check started at the target stationary law (log-uniform at $q = 1$). It did not establish relaxation and is excluded here.

8 Conclusion

Uniform microscopic activity can give channels coupled to more interactions stronger state fluctuations, without reference to a particular spatial geometry. Under the specified conserved-flow dynamics, their stationary residence times yield an exact concentration window and a measurable finite-population correction. The upper edge limits extreme pooled concentration; it does not forbid all large channels or unequal sharing. The proposed physical test concerns the interaction mechanism, its conditional variance law and the resulting finite-range curve. It is not a universal amplitude window, a theorem of channel formation or a derivation of biological selection.

Code and reproducibility. Sources, tests, retained data and reproduction commands are at <https://github.com/divadwg/selection-as-physics>; see `paper/README.md`.

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